

Modeling Bayesian Transport Map Uncertainty for Non-Gaussian Spatial Data via Laplace Approximation

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Motivation: Modeling Non-Gaussian Spatial Fields



Non-Gaussian spatial fields can arise in many scientific applications:

- Environmental Data [Fonseca et al., 2023]
- Climate Model Emulation [Chakraborty and Katzfuss, 2025]
- Medical Imaging Data [Gopinath et al., 2018]
- Disease Incidence [Kang et al., 2024]

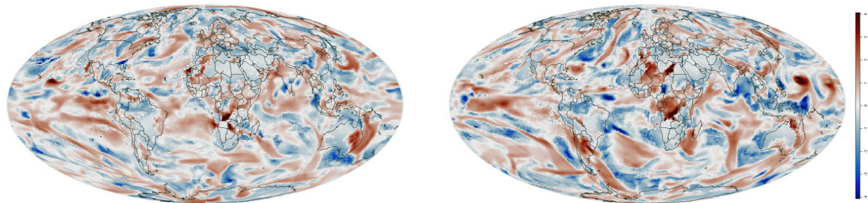


Figure: Log-transformed precipitation anomalies from a climate model

Let $\mathbf{y} = (y_1, \dots, y_N)^\top$ denote an N -dimensional random vector (spatial field).

If $\mathbf{y} \sim \mathcal{N}_N(\mathbf{0}, \mathbf{\Sigma})$ we can represent $\mathbf{\Sigma}^{-1} = \mathbf{L}^\top \mathbf{L}$.

The (lower-triangular) Cholesky factor $\mathbf{L}$ transforms $\mathbf{y}$ to a standard multivariate normal:

$$\mathbf{z} = \mathbf{L}\mathbf{y} \sim \mathcal{N}_N(\mathbf{0}, \mathbf{I}_N)$$

Cholesky Factors

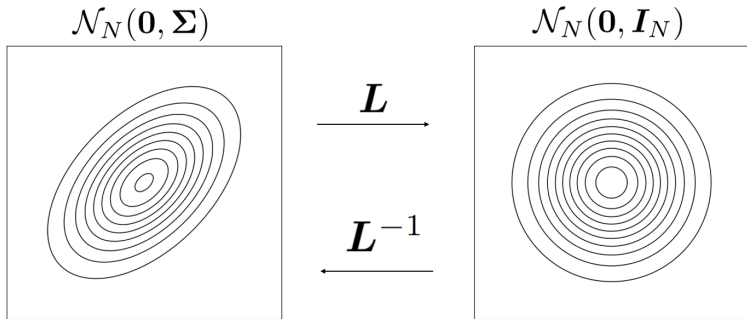


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Let $\mathbf{y} = (y_1, \dots, y_N)^\top$ denote an N -dimensional random vector (spatial field) with joint (non-Gaussian) distribution $p(\mathbf{y})$, centered to have mean $\mathbf{0}$.

A *lower-triangular transport map* $\mathcal{T} : \mathbb{R}^N \rightarrow \mathbb{R}^N$ also maps $p(\mathbf{y})$ to a standard multivariate normal:

$$\mathbf{z} = \mathcal{T}(\mathbf{y}) \sim \mathcal{N}_N(\mathbf{0}, I_N) \quad \text{if} \quad \mathbf{y} \sim p(\mathbf{y}),$$

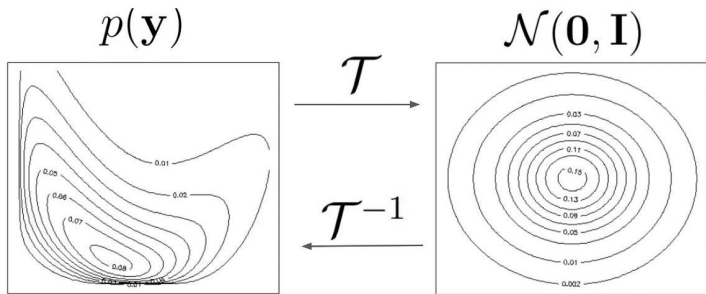
Transport Maps



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Conditional Factorization:

$$p(\mathbf{y}) = p_z(\mathcal{T}(\mathbf{y})) |\det \nabla \mathcal{T}|$$
$$\prod_{n=1}^N p(y_n | \mathbf{y}_{1:n}) = \prod_{n=1}^N \left(p_z(\mathcal{T}_n(\mathbf{y}_{1:n})) \left| \frac{\partial \mathcal{T}_n(\mathbf{y}_{1:n})}{\partial y_n} \right| \right)$$

We assume that:

$$\mathcal{T}_n(\mathbf{y}_{1:n}) = \frac{y_n - f_n(\mathbf{y}_{1:n-1})}{d_n},$$

where $f_n : \mathbb{R}^{n-1} \rightarrow \mathbb{R}$ is a regression function, and $d_n > 0$ is a conditional sd.

$$\prod_{n=1}^N p(y_n | \mathbf{y}_{1:n}) = \prod_{n=1}^N \left(p_z(\mathcal{T}_n(\mathbf{y}_{1:n})) \left| \frac{\partial \mathcal{T}_n(\mathbf{y}_{1:n})}{\partial y_n} \right| \right) = \prod_{n=1}^N \mathcal{N}(y_n | f_n(\mathbf{y}_{1:n-1}), d_n^2)$$

If we assume conditional independence in $p(\mathbf{y})$ we induce sparsity in the map $\mathcal{T}$:

$$f_n(\mathbf{y}_{1:n-1}) \approx f_n(y_{c(n)}), \quad c(n) \subset \{1, \dots, n-1\}, \quad |c(n)| = m$$

The transport map components are modeled as follows :

$$\begin{aligned}d_n^2 &\sim \text{Inverse-Gamma}(\alpha_n, \beta_n) \\ f_n \mid d_n &\sim \mathcal{GP}(0, d_n^2 K_n(\cdot, \cdot))\end{aligned}$$

where $K_n(\cdot, \cdot)$ is a chosen covariance/kernel function (e.g., Matérn).

With priors of this form, and some additional assumptions (See Katzfuss and Schäfer [2024] for full details), the transport map structure relies on

$$\boldsymbol{\theta} = (\theta_{\sigma_1}, \theta_{\sigma_2}, \theta_{d_1}, \theta_{d_2}, \theta_{\gamma}, \theta_q)^\top$$

- θ_q controls the sparsity of $\mathcal{T}$.
- θ_{σ_1} and θ_{σ_2} control the degree of nonlinearity.

Past work on this model¹ has taken an empirical Bayesian approach and fixed θ at the value $\hat{\theta}$ that maximizes $\log p_{\theta}(\mathbf{y})$.

Because $\log p_{\theta}(\mathbf{y})$ is a sum of N relatively simple terms, it is straightforward to optimize using gradient descent.

¹Chakraborty and Katzfuss [2025], Katzfuss and Schäfer [2024], Wiemann and Katzfuss [2023]

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By fixing θ we are ignoring any variability or uncertainty in the map itself.

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Can we understand and account for the variability/uncertainty in map hyperparameters in an efficient way?

To assess the amount of uncertainty in the transport map hyperparameters we compare three methods:

- Empirical Bayesian model using gradient descent
- Approximate the posterior distribution of θ using a Laplace approximation
- Fully Bayesian model using MCMC

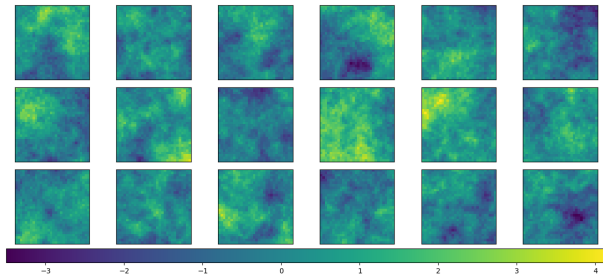
Let $\mathcal{L}(\boldsymbol{\theta}) = \log p_{\boldsymbol{\theta}}(\mathbf{y})$ represent the log-likelihood of the transport map model as a function of $\boldsymbol{\theta}$.

After obtaining $\hat{\boldsymbol{\theta}}$ via gradient descent, we calculate $H = -\nabla_{\boldsymbol{\theta}}^2 L(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}$.

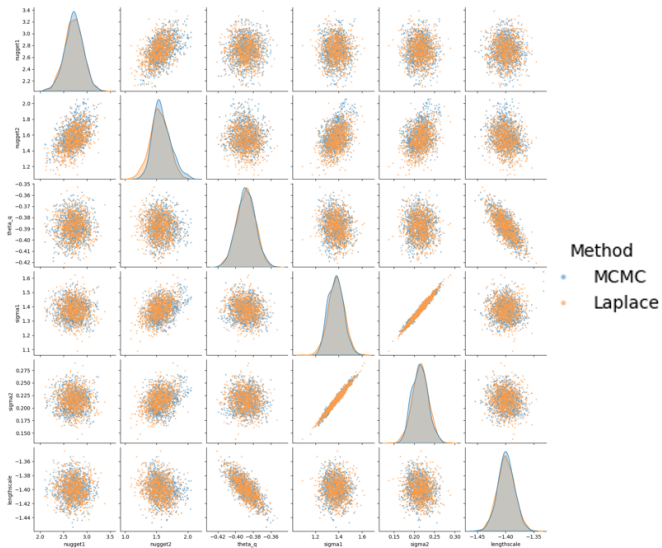
$$p(\boldsymbol{\theta}|\mathbf{y}) \approx \hat{p}(\boldsymbol{\theta}|\mathbf{y}) = \mathcal{N}(\hat{\boldsymbol{\theta}}, H^{-1})$$

If $p(\boldsymbol{\theta}|\mathbf{y})$ is close to Gaussian, then $\hat{p}(\boldsymbol{\theta}|\mathbf{y})$ should be a good approximation to the true posterior.

We compared the posterior distributions of θ ($p(\theta|\mathbf{y})$) and the posterior predictive distributions of $\mathbf{y}^*$ ($p(\mathbf{y}^*|\mathbf{y})$) under each of the models using simulated data while varying the number of training samples/fields used to fit the model.

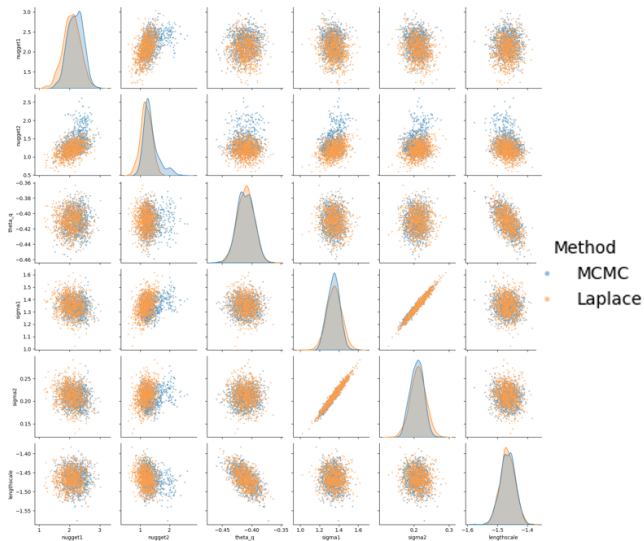


Simulation Study Results



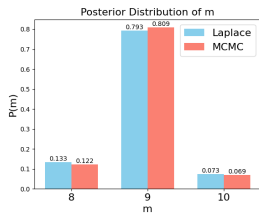
nTrain = 100

Simulation Study Results

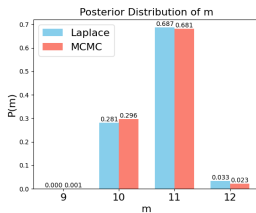


nTrain = 50

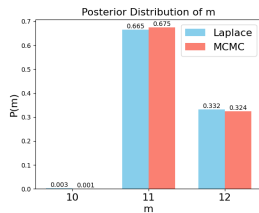
Transport Map Sparsity: m



$n_{\text{Train}} = 20$

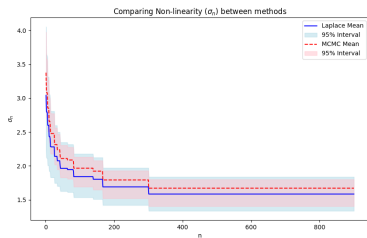


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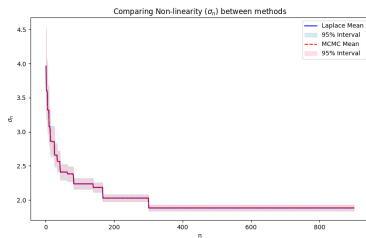


$n_{\text{Train}} = 100$

Transport Map Nonlinearity: σ_n



nTrain = 20



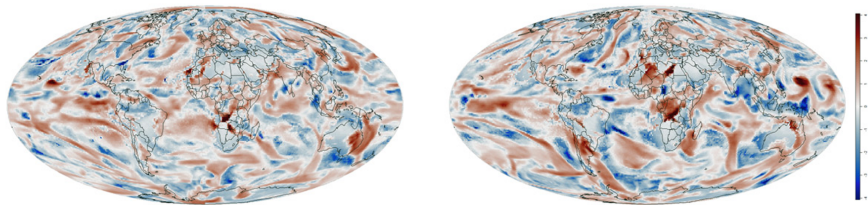
nTrain = 100

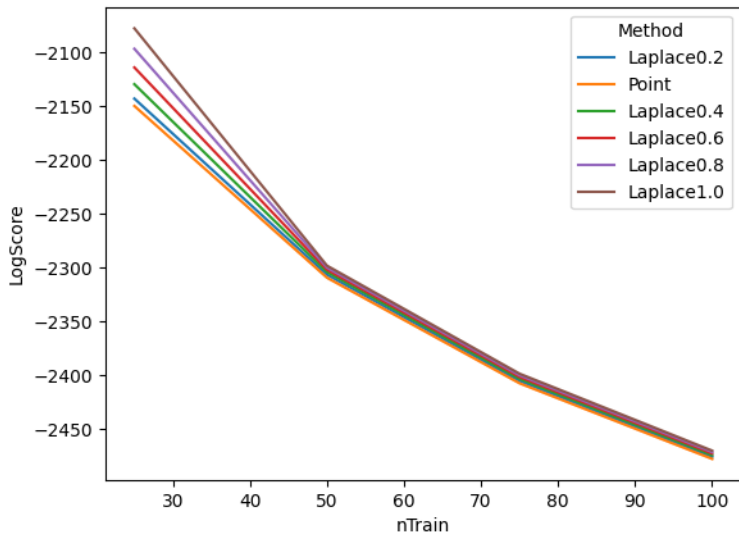


It appears that the Laplace Approximation is valid when we use a large enough training set.

We are able to capture and understand the uncertainty in Transport Map structure; however, we also want to evaluate if including the uncertainty in the map structure actually improves model fit.

To evaluate if modeling the uncertainty in the model improved model fit, we use our methods to model log-transformed precipitation anomalies from a climate model with different sized training sets. We then evaluated the model fit using log-score.





In conclusion, in this work we considered 3 models for transport map uncertainty

- Empirical Bayesian model using gradient descent
- Approximate the posterior distribution of θ using a Laplace approximation
- Fully Bayesian model using MCMC

With a large enough set of training samples, a Laplace Approximation for the posterior of map hyperparameters is reasonable: $p(\theta|\mathbf{y}) \approx \hat{p}(\theta|\mathbf{y}) = \mathcal{N}(\hat{\theta}, H^{-1})$.

We were able to measure the uncertainty in hyperparameters and their impact on the transport map structure (sparsity and non-linearity).

In practice, this did not lead to significant changes in model fit to real-world data.

Questions?



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